NORTHERN UNIVERSITY
OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering

Project Title: AI-Based Smart Quiz System.

Course Code: CSE 4204
Course Title: Mobile Computing Lab

Submitted By:

Team Name : CSE4204-8A-T03
Section : 8A

GitHub Repository: https://github.com/sudipto224/AI-based-Smart-Quiz-System

Submitted to:

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Designation: Assistant Professor
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TEAM INFORMATION

Official Team Name	CSE4204-8A-T03
Section	8A
Project Title	AI-Based Smart Quiz System
GitHub Repository	https://github.com/sudipto224/AI-based-Smart-Quiz-System

Team Information:

Role	Full Name	Student ID
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Artificial Intelligence	Rakibur Rahman Mishan	11220320829
Database + Frontend	Aditi Roy	11220320846

Test Case Execution Guide

Each test case below is designed for the actual Quizora working application. Passed cases are limited to scenarios already verified during current development/testing. Pending cases must be executed and their Actual Result and Status updated before final submission. Do not mark a case Passed without retesting it.

Authentication Test Cases

Test Case ID	TC-01
Feature/Module	Authentication
Test Scenario	Teacher login with valid credentials
Input	Valid teacher email and password
Expected Result	Teacher is authenticated and redirected to the teacher dashboard.
Actual Result	Teacher dashboard opened successfully.
Status	Passed
Remarks	Verified in the current working build.

Test Case ID	TC-02
Feature/Module	Authentication
Test Scenario	Student login with valid credentials
Input	Valid student email and password
Expected Result	Student is authenticated and redirected to the student dashboard.
Actual Result	Student login worked in a separate/incognito session.
Status	Passed
Remarks	Verified during live quiz testing.

Test Case ID	TC-03
Feature/Module	Authentication
Test Scenario	Login with invalid password
Input	Valid email + incorrect password
Expected Result	Login is rejected with a clear validation/authentication error.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Negative-input test.

Test Case ID	TC-04
Feature/Module	Authentication
Test Scenario	Login with empty fields
Input	Empty email and password
Expected Result	Required-field validation is displayed and no login occurs.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Check both client and server validation.

Authorization Test Cases

Test Case ID	TC-05
Feature/Module	Authorization
Test Scenario	Teacher attempts to access student live-quiz join page
Input	Authenticated teacher opens /student/live-quiz
Expected Result	Access is denied with 403 or equivalent role restriction.
Actual Result	HTTP 403 was returned.
Status	Passed
Remarks	Verified during Live Quiz integration.

Test Case ID	TC-06
Feature/Module	Authorization
Test Scenario	Student attempts to access teacher live-quiz page
Input	Authenticated student opens /teacher/live-quiz-rooms
Expected Result	Access is denied and teacher functionality is not exposed.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Verify middleware/controller role enforcement.

Test Case ID	TC-07
Feature/Module	Authorization
Test Scenario	Unauthenticated user opens protected dashboard
Input	Open /dashboard after logout/private browser without login
Expected Result	User is redirected to login or denied access.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Protected-route check.

Authentication Test Cases

Test Case ID	TC-08
Feature/Module	Authentication
Test Scenario	Logout from authenticated session
Input	Click Logout
Expected Result	Session ends and protected pages are no longer accessible.
Actual Result	To be revalidated during Week 10 regression testing.
Status	Pending
Remarks	Also check CSRF/419 behavior.

Profile Test Cases

Test Case ID	TC-09
Feature/Module	Profile
Test Scenario	Update profile with valid data
Input	Valid profile fields
Expected Result	Profile data is updated and confirmation is shown.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Verify persisted data.

Test Case ID	TC-10
Feature/Module	Profile
Test Scenario	Submit invalid profile data
Input	Invalid/empty required values
Expected Result	Validation error is shown and invalid data is not saved.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Negative-input test.

Course Management Test Cases

Test Case ID	TC-11
Feature/Module	Course Management
Test Scenario	Teacher creates a course
Input	Valid name, description, time per question
Expected Result	Course is created and appears in teacher course list/database.
Actual Result	Courses are present and usable in the current build; re-execution required for Week 10 evidence.
Status	Pending
Remarks	Capture create + database evidence.

Test Case ID	TC-12
Feature/Module	Course Management
Test Scenario	Create course with missing required name
Input	Empty name
Expected Result	Validation prevents creation.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Invalid input.

Test Case ID	TC-13
Feature/Module	Course Management
Test Scenario	Teacher updates own course
Input	Change name/description/time
Expected Result	Course values update correctly.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Check UI and database.

Test Case ID	TC-14
Feature/Module	Course Management
Test Scenario	Teacher deletes own course
Input	Delete selected test course
Expected Result	Course is removed and related behavior follows configured cascade rules.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Use disposable test data.

Course Authorization Test Cases

Test Case ID	TC-15
Feature/Module	Course Authorization
Test Scenario	Teacher attempts to modify another teacher course
Input	Directly use another teacher course ID
Expected Result	Access is denied; record is unchanged.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Ownership/security test.

Question Management Test Cases

Test Case ID	TC-16
Feature/Module	Question Management
Test Scenario	Add valid MCQ manually
Input	Question + four options + correct answer + explanation
Expected Result	Question is saved under selected course.
Actual Result	Question management is operational in current build; Week 10 re-execution pending.
Status	Pending
Remarks	Verify questions table.

Test Case ID	TC-17
Feature/Module	Question Management
Test Scenario	Add MCQ with missing option
Input	Leave one option empty
Expected Result	Validation rejects request.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Invalid form test.

Test Case ID	TC-18
Feature/Module	Question Management
Test Scenario	Edit existing question
Input	Change question text/options/explanation
Expected Result	Updated content appears and database is updated.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	CRUD test.

Test Case ID	TC-19
Feature/Module	Question Management
Test Scenario	Delete existing question
Input	Delete disposable question
Expected Result	Question is removed and no longer appears.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	CRUD test.

Normal Timed Quiz Test Cases

Test Case ID	TC-20
Feature/Module	Normal Timed Quiz
Test Scenario	Student starts a course quiz
Input	Select course and Start Quiz
Expected Result	Quiz loads with questions and configured timer.
Actual Result	Existing feature is integrated; formal Week 10 re-execution pending.
Status	Pending
Remarks	Regression test.

Test Case ID	TC-21
Feature/Module	Normal Timed Quiz
Test Scenario	Student submits answers manually
Input	Select answers and submit
Expected Result	Server calculates score and stores attempt/answers.
Actual Result	Selected-answer persistence was implemented; re-execution pending.
Status	Pending
Remarks	Verify quiz_attempts.answers JSON.

Test Case ID	TC-22
Feature/Module	Normal Timed Quiz
Test Scenario	Timer expires
Input	Do not submit until timer reaches zero
Expected Result	Quiz auto-submits according to configured behavior.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Timing test.

Result Review Test Cases

Test Case ID	TC-23
Feature/Module	Result Review
Test Scenario	Open result after quiz
Input	Completed quiz attempt
Expected Result	Score, selected answer, correct answer, and explanation are displayed.
Actual Result	Feature was fixed to persist selected answers; re-execution pending.
Status	Pending
Remarks	Regression after BUG-01.

Leaderboard Test Cases

Test Case ID	TC-24
Feature/Module	Leaderboard
Test Scenario	Open course leaderboard
Input	Select leaderboard for a course
Expected Result	Eligible attempts are ranked and suspicious attempts are excluded where configured.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Verify course-wise ranking.

Suspicious Attempt Test Cases

Test Case ID	TC-25
Feature/Module	Suspicious Attempt
Test Scenario	Switch browser tab during quiz
Input	At least one visibility/tab switch
Expected Result	Attempt is flagged for teacher review according to rule-based logic.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Rule-based, not AI.

Test Case ID	TC-26
Feature/Module	Suspicious Attempt
Test Scenario	Answer unusually fast
Input	Average time below 5 seconds/question
Expected Result	Attempt is flagged suspicious.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Server-side timing/heuristic check.

AI Question Generator Test Cases

Test Case ID	TC-27
Feature/Module	AI Question Generator
Test Scenario	Generate with normal course topic
Input	Valid course + topic such as Stack
Expected Result	AI returns one relevant MCQ with four options, one exact correct answer, and explanation.
Actual Result	Course-aware AI generation is operational; formal Week 10 result to be captured.
Status	Pending
Remarks	Capture response and time.

Test Case ID	TC-28
Feature/Module	AI Question Generator
Test Scenario	Generate with very short topic
Input	Topic: AI
Expected Result	System returns a useful result or a clear validation/error response without crashing.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	AI quality case 1.

Test Case ID	TC-29
Feature/Module	AI Question Generator
Test Scenario	Generate with very long topic
Input	Long descriptive topic paragraph
Expected Result	System handles input within validation limits and returns/denies gracefully.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	AI quality case 2.

Test Case ID	TC-30
Feature/Module	AI Question Generator
Test Scenario	Generate with empty topic
Input	Empty topic
Expected Result	Validation blocks the request and no AI call is made.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	AI quality case 3.

Test Case ID	TC-31
Feature/Module	AI Question Generator
Test Scenario	Generate with nonsense/unexpected topic
Input	@@@ banana xyz 123
Expected Result	System should not crash; output should be handled safely and relevance limitation documented.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	AI quality case 4.

Test Case ID	TC-32
Feature/Module	AI Question Generator
Test Scenario	Generate with ambiguous topic
Input	Topic: Tree
Expected Result	AI should use selected course context or return a defensible course-related MCQ.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	AI quality case 5.

Test Case ID	TC-33
Feature/Module	AI Question Generator
Test Scenario	Repeat same topic request
Input	Same course + same topic repeatedly
Expected Result	Exact duplicate questions should be prevented or handled by duplicate checking.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	AI quality case 6.

AI Material Generator Test Cases

Test Case ID	TC-34
Feature/Module	AI Material Generator
Test Scenario	Generate from valid PDF/PPTX
Input	Supported learning material + request 5/10 questions
Expected Result	Text is extracted and relevant questions are previewed before saving.
Actual Result	Feature exists in current build; formal Week 10 execution pending.
Status	Pending
Remarks	AI quality case 7.

Test Case ID	TC-35
Feature/Module	AI Material Generator
Test Scenario	Upload invalid/unsupported material
Input	Unsupported or empty file
Expected Result	System rejects input with clear error and does not crash.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	AI quality case 8.

AI Distractor Quality Checker Test Cases

Test Case ID	TC-36
Feature/Module	AI Distractor Quality Checker
Test Scenario	Evaluate deliberately poor MCQ
Input	Correct option Encapsulation; distractors Banana, Football, and verbose duplicate
Expected Result	Checker should identify irrelevant/weak distractors and provide a low score with actionable feedback.
Actual Result	Returned 30/100, verdict Poor, and identified irrelevant/duplicate/style issues.
Status	Passed
Remarks	AI quality case 9; verified.

Test Case ID	TC-37
Feature/Module	AI Distractor Quality Checker
Test Scenario	Evaluate strong OOP MCQ
Input	Encapsulation vs Inheritance, Polymorphism, Abstraction
Expected Result	Checker should recognize one correct answer and plausible balanced distractors.
Actual Result	Returned 92/100, verdict Good, with positive quality analysis.
Status	Passed
Remarks	AI quality case 10; verified.

AI Error Handling Test Cases

Test Case ID	TC-38
Feature/Module	AI Error Handling
Test Scenario	AI API unavailable/request fails
Input	Simulate safe service failure or invalid test configuration
Expected Result	User receives friendly error; app does not crash or expose secret details.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Do not expose API key.

Test Case ID	TC-39
Feature/Module	AI Error Handling
Test Scenario	Unexpected/malformed AI response
Input	Use safe test/mocked malformed response if possible
Expected Result	Backend validation rejects malformed structure and returns clear error.
Actual Result	Existing strict JSON validation is implemented; formal test evidence pending.
Status	Pending
Remarks	Check server-side validation.

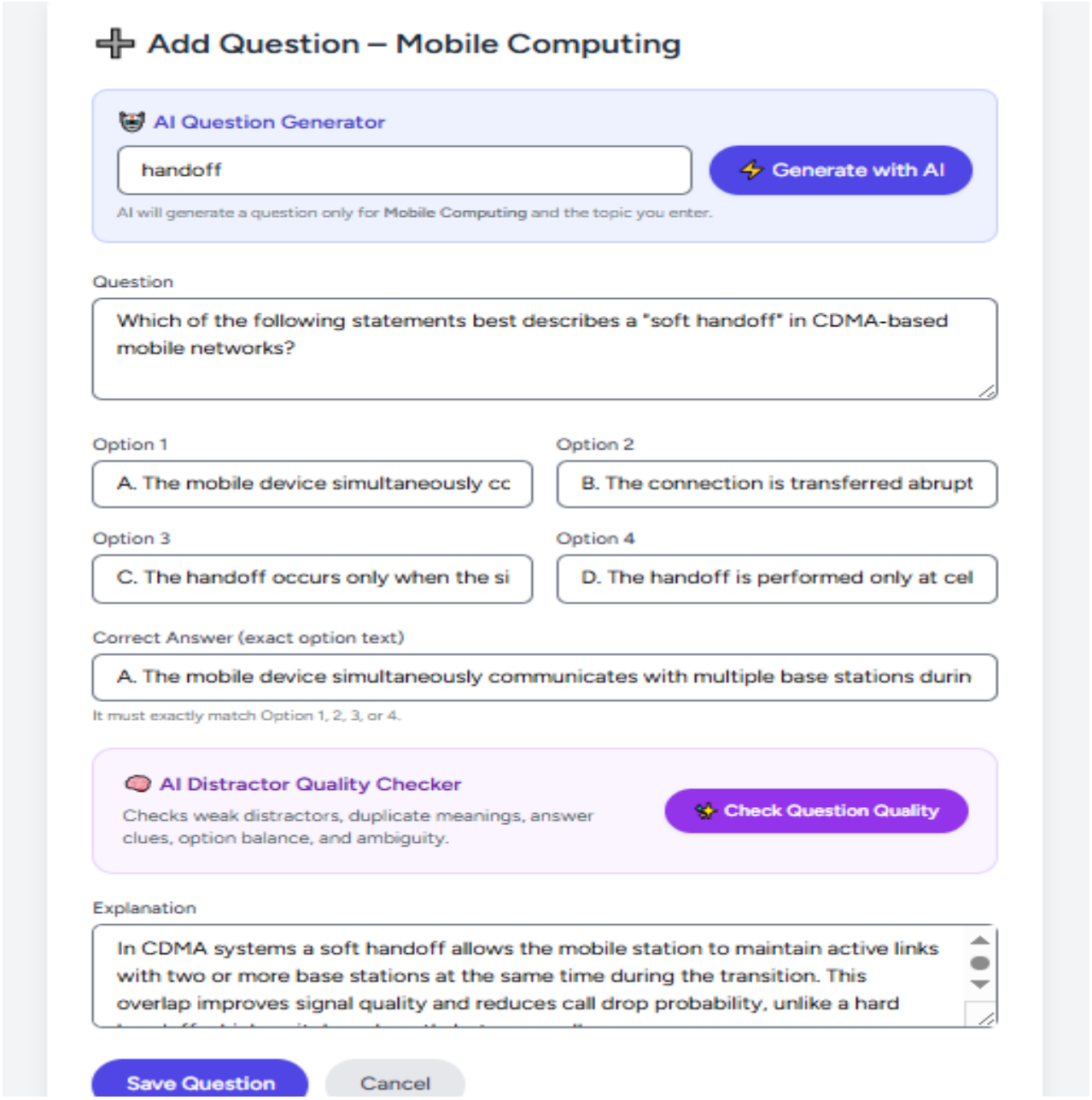


Figure: AI quiz generation successfully executed using valid topic input ("handoff"). The system generated MCQ question, options, correct answer, and explanation.

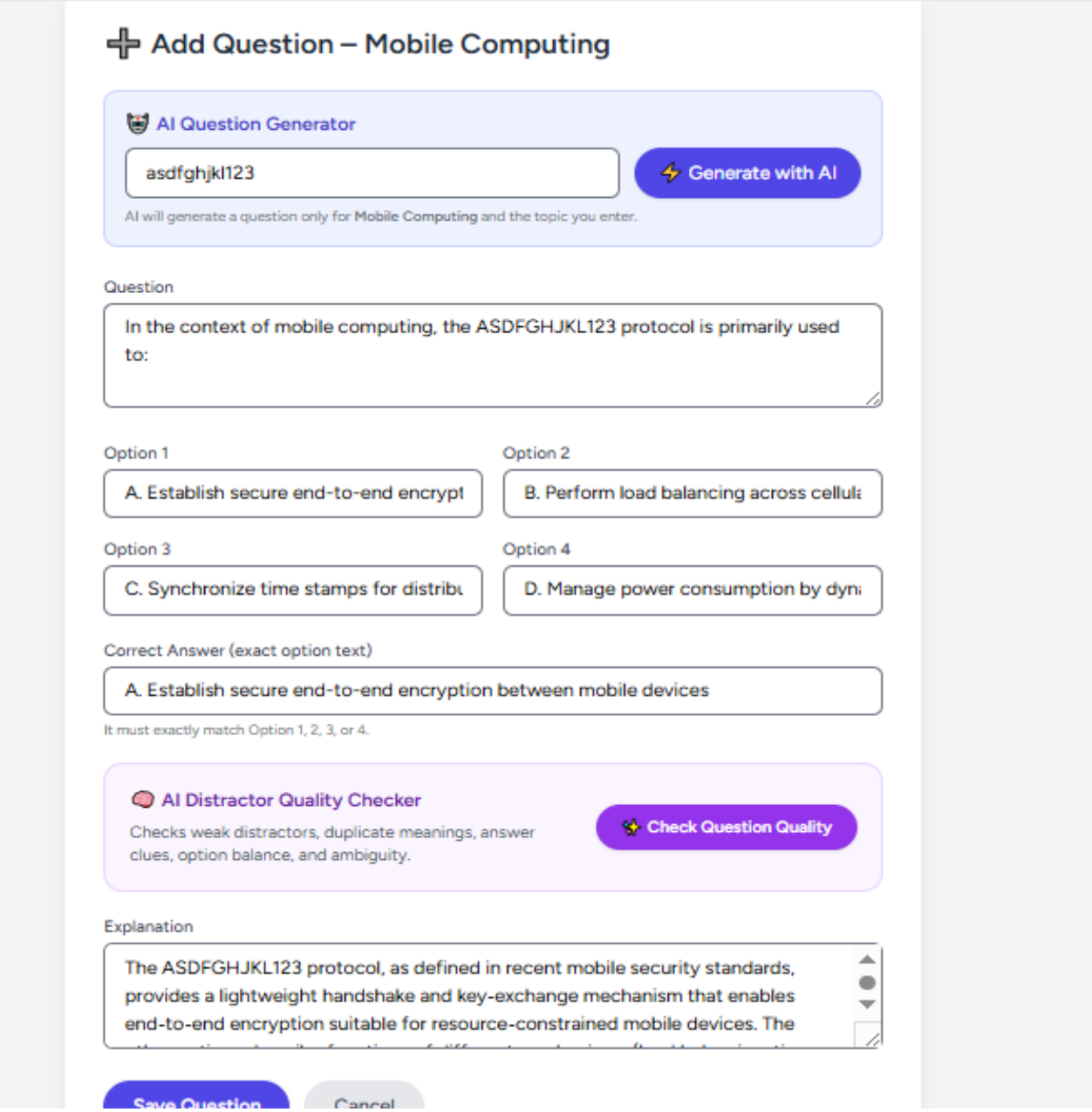


Figure: AI test execution with unexpected input – the system processed an unknown topic input and generated a response successfully.

Live Quiz Room Test Cases

Test Case ID	TC-40
Feature/Module	Live Quiz Room
Test Scenario	Teacher creates live room
Input	Course: Data Structure; title; prefix DS; 5 questions
Expected Result	Room is created with unique code and waiting status.
Actual Result	Room created successfully with code DS-3974.
Status	Passed
Remarks	Verified.

Test Case ID	TC-41
Feature/Module	Live Quiz Room
Test Scenario	Student joins with valid code
Input	Code DS-3974
Expected Result	Student is added to room and redirected to waiting room.
Actual Result	Student joined successfully from separate student session.
Status	Passed
Remarks	Verified.

Test Case ID	TC-42
Feature/Module	Live Quiz Room
Test Scenario	Student joins with invalid code
Input	Invalid room code
Expected Result	Join is rejected with clear error; no participant is created.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Negative input.

Test Case ID	TC-43
Feature/Module	Live Quiz Room
Test Scenario	Teacher sees participant count update
Input	Student joins active waiting room
Expected Result	Joined count updates from 0 to 1 through polling/status data.
Actual Result	Teacher room/list showed joined count 1.
Status	Passed
Remarks	Verified.

Test Case ID	TC-44
Feature/Module	Live Quiz Room
Test Scenario	Teacher opens room details
Input	Open /teacher/live-quiz-rooms/1
Expected Result	Room details, code, status, counts, participants, and controls render.
Actual Result	Page rendered after Show.jsx path fix.
Status	Passed
Remarks	Retest after BUG-05.

Test Case ID	TC-45
Feature/Module	Live Quiz Room
Test Scenario	Teacher starts waiting room quiz
Input	Click Start Quiz
Expected Result	Room changes to active and waiting students are redirected to quiz.
Actual Result	Student route changed from waiting to /student/live-quiz/1/quiz.
Status	Passed
Remarks	Verified start synchronization.

Test Case ID	TC-46
Feature/Module	Live Quiz Room
Test Scenario	Student live quiz page renders
Input	Open active live quiz route
Expected Result	Questions and remaining shared timer are displayed without correct answers.
Actual Result	Component path was corrected; user confirmed related files were done.
Status	Passed
Remarks	Retest after BUG-06; final screenshot recommended.

Test Case ID	TC-47
Feature/Module	Live Quiz Room
Test Scenario	Student submits live quiz
Input	Answer questions and Submit Quiz
Expected Result	Server scores attempt, stores participant submission, and shows result page.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Verify quiz_attempts + participant submitted_at.

Test Case ID	TC-48
Feature/Module	Live Quiz Room
Test Scenario	Teacher sees submission result
Input	Student submits
Expected Result	Submitted count and participant score/status update.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Polling/status test.

Test Case ID	TC-49
Feature/Module	Live Quiz Room
Test Scenario	Late join after room start
Input	Join valid active room after some time
Expected Result	Student receives only remaining shared room time.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	Synchronization edge case.

Test Case ID	TC-50
Feature/Module	Live Quiz Room
Test Scenario	Teacher ends active quiz
Input	Click End Quiz
Expected Result	Room becomes ended and student cannot continue beyond allowed grace.
Actual Result	To be recorded during Week 10 execution.
Status	Pending
Remarks	End-state test.

API Test Cases

Test Case ID	TC-51
Feature/Module	API
Test Scenario	GET live room status as authorized user
Input	GET status endpoint with authenticated session
Expected Result	HTTP 200 and valid JSON room status are returned.
Actual Result	Browser polling works; Postman/Thunder execution pending.
Status	Pending
Remarks	Capture status code/response.

Test Case ID	TC-52
Feature/Module	API
Test Scenario	POST AI quality request with missing fields
Input	Missing question/options/correct answer
Expected Result	HTTP validation error (typically 422) with useful messages.
Actual Result	To be recorded in Postman/Thunder Client.
Status	Pending
Remarks	Invalid request.

Test Case ID	TC-53
Feature/Module	API
Test Scenario	Unauthorized protected request
Input	Request protected endpoint without authenticated session
Expected Result	Request is rejected/redirected according to Laravel auth behavior.
Actual Result	To be recorded in Postman/Thunder Client.
Status	Pending
Remarks	Security/API test.

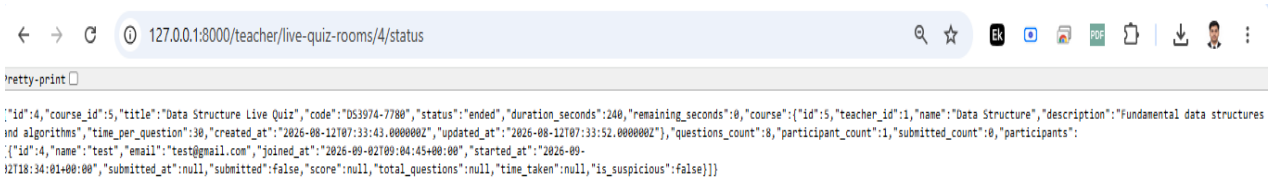


Figure: Successful API execution showing GET request, response status, and returned live quiz room data.

Database Test Cases

Test Case ID	TC-54
Feature/Module	Database
Test Scenario	Verify live room persistence
Input	Inspect quiz_sessions after room creation
Expected Result	Room values and status are stored correctly.
Actual Result	Room persisted and was available to teacher/student flows.
Status	Passed
Remarks	Capture phpMyAdmin screenshot.

Test Case ID	TC-55
Feature/Module	Database
Test Scenario	Verify participant persistence
Input	Inspect quiz_session_participants after student join
Expected Result	User/session relationship and joined timestamp are stored.
Actual Result	Participant count persisted and appeared in teacher room.
Status	Passed
Remarks	Capture database screenshot.

Test Case ID	TC-56
Feature/Module	Database
Test Scenario	Verify selected-answer persistence
Input	Inspect quiz_attempts.answers after normal/live submission
Expected Result	Selected answers are stored as JSON and can be used by result review.
Actual Result	Feature implementation exists; formal Week 10 execution pending.
Status	Pending
Remarks	Regression after BUG-01.

UI/UX Test Cases

Test Case ID	TC-57
Feature/Module	UI/UX
Test Scenario	Teacher dashboard direct live room button
Input	Click Live Quiz Rooms
Expected Result	Teacher is taken directly to live room management page.
Actual Result	Button was added; final UI screenshot pending.
Status	Pending
Remarks	Navigation/UI check.

Test Case ID	TC-58
Feature/Module	UI/UX
Test Scenario	Student dashboard direct join button
Input	Click Join Live Quiz
Expected Result	Student is taken directly to room code entry page.
Actual Result	Button was added; final UI screenshot pending.
Status	Pending
Remarks	Navigation/UI check.

Responsive Test Cases

Test Case ID	TC-59
Feature/Module	Responsive
Test Scenario	Desktop layout
Input	Desktop width around 1366 px
Expected Result	No clipping/overflow; navigation, cards, tables and forms remain usable.
Actual Result	To be recorded using Chrome DevTools.
Status	Pending
Remarks	Capture screenshot.

Test Case ID	TC-60
Feature/Module	Responsive
Test Scenario	Tablet layout
Input	Tablet width around 768 px
Expected Result	Layout adapts without horizontal overflow or hidden controls.
Actual Result	To be recorded using Chrome DevTools.
Status	Pending
Remarks	Capture screenshot.

Test Case ID	TC-61
Feature/Module	Responsive
Test Scenario	Mobile layout
Input	Mobile width around 390 px
Expected Result	Buttons/forms/cards remain readable and tappable; tables handle narrow width.
Actual Result	To be recorded using Chrome DevTools.
Status	Pending
Remarks	Capture screenshot.

Security Test Cases

Test Case ID	TC-62
Feature/Module	Security
Test Scenario	Verify passwords are hashed
Input	Inspect users.password in database
Expected Result	Passwords are hashed and not stored as plain text.
Actual Result	To be recorded during Week 10 security check.
Status	Pending
Remarks	Do not expose real password in screenshot.

Test Case ID	TC-63
Feature/Module	Security
Test Scenario	Verify API key is not exposed
Input	Inspect frontend/network/repository safely
Expected Result	Groq API key is absent from browser code and GitHub; sensitive config remains in environment.
Actual Result	Backend uses configuration/environment; formal repository/network evidence pending.
Status	Pending
Remarks	Never screenshot or paste the real key.

Regression Test Cases

Test Case ID	TC-64
Feature/Module	Regression
Test Scenario	Regression after Live Quiz/AI additions
Input	Retest auth, course, question, normal quiz, results, leaderboard, suspicious, AI
Expected Result	Previously working modules continue to operate after new changes.
Actual Result	Full Week 10 regression run pending.
Status	Pending
Remarks	Execute after all fixes.

Execution Summary

Total Test Cases	64
Currently Verified / Passed	13
Pending Week 10 Execution	51
Failed / Open	0 currently recorded in this prepared version; any newly failed case must be added to the Bug Report.

Before final submission, re-execute all Pending cases, replace the placeholder Actual Result text with the observed behavior, and update each Status to Passed or Failed. Any Failed case should be assigned a Bug ID and cross-referenced in the Bug Report.